

DARSHAN R

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EXECUTIVE SUMMARY

Final-year B.E. Computer Science and Engineering student specializing in AI/ML, Generative AI, agentic systems, and computer vision with Project Management skills. Experienced in building end-to-end AI applications involving LLMs, RAG pipelines, multi-agent workflows, and real-time vision systems. Strong Python and software engineering foundation with hands-on experience developing and evaluating production-grade AI solutions.

EDUCATIONAL QUALIFICATION

Sri Ramakrishna Engineering College, Coimbatore - B.E Computer Science Engineering CGPA: 7.08 (till 6th semester)	2023 - 2027
Shree Sarasswathi Vidhyaah Mandheer (SSVM), Coimbatore – HSC (CBSE) Percentage: 72.4%	2022 - 2023
Adithya International School (AIS), Coimbatore – SSLC (CBSE) Percentage: 72%	2020 - 2021

EXPERIENCE

AI/ML Intern – Nxtlogic Software Solutions

June 2025

- Developed a Used Car Price Estimation System using machine learning algorithms to predict vehicle prices based on features such as mileage, brand, fuel type, and year.
- Conducted data preprocessing, feature engineering, and model selection to improve prediction accuracy, enhance model robustness, and ensure consistent performance across real-world scenarios.
- Implemented regression models using Python and optimized performance through hyperparameter tuning.
- Applied Explainable AI techniques (SHAP) to interpret model predictions and identified key factors influencing pricing.
- Built an end-to-end ML workflow including data cleaning, model training, evaluation, and result visualization.

UI/UX Intern – VisionCorp InfoTech

December 2025

- Designed a professional-grade UI/UX for a cab/ride booking system, focusing on usability, accessibility and cognitive design principles to build rider trust, drive engagement, and improve overall app retention rates.
- Conducted in-depth user research and analysis to understand customer behavior, expectations, and pain points.
- Created user flows, wireframes, and high-fidelity UI designs for seamless user interaction.
- Improved user experience by optimizing navigation, booking flow, and interface clarity.
- Collaborated with stakeholders to ensure designs aligned with real-world user needs and business goals.

PROJECTS

Agentic Supermarket POS Architecture

- Built a unified mobile and web POS with AI chat interface for checkout, inventory, billing, and store operations.
- Designed an agentic architecture with domain skills, typed tools, multi-turn state, and persistent memory.
- Integrated AI barcode scanning, thermal printing, FEFO inventory, GST billing, and multilingual voice/text interaction.
- Engineered a secure multi-tenant architecture with RBAC and automated backups, validated via 192 tests.
- Developed a CRM analytics engine tracking lifetime customer spend, visit counts, and purchase patterns.

Enterprise AI Recruitment & Proctoring Engine

- Architected a gateway isolating WebRTC signaling from CPU-bound ML pipelines for minimal latency execution.
- Engineered an evaluation pipeline using dense-vector embeddings and NLP for high-precision latent space ranking.

- Developed a privacy-focused biometric proctoring engine processing facial meshes for real-time anomaly detection.
- Implemented a testing framework to continually monitor active Socket.IO connections and memory footprints.
- Integrated sovereign, on-premise LLMs featuring strict schema constraints to mitigate prompt injection attacks, utilizing resilient heuristic fallbacks to generate deterministic analytics with high data privacy.

Robust Vision Pipeline for Real-Time Surveillance

- Architected a real-time deep learning vision system for intelligent object recognition and anomaly detection.
- Designed evaluation benchmarks to test model against noisy inputs commonly found in real-world environments.
- Reduced false-positive alerts via failure analysis, systematic experimentation, and synthetic data augmentation.
- Optimized inference latency and accuracy across the deployment pipeline for continuous, high-fidelity monitoring.

Multi-Agent Equity Research & Portfolio Intelligence

- Orchestrated a 15-node LangGraph multi-agent LLM ecosystem for autonomous equity market intelligence.
- Automated a parallel NIFTY 50 market scanner utilizing semantic caching and multi-model inference.
- Enforced paper-trading risk controls via ATR position sizing, AI confidence scoring, and circuit breakers.
- Embedded MCP tooling and a BM25-indexed reflection loop for adaptive, memory-augmented simulated execution.

Scalable RAG System for Document Intelligence

- Built a layout-aware ingestion pipeline to parse heterogeneous, unstructured documents at scale.
- Engineered ChromaDB-backed semantic retrieval for schema-driven structured entity extraction.
- Deployed local open-weight LLMs for private, on-premise document understanding with less API dependency.
- Implemented evidence-grounded verification and ground-truth benchmarking to suppress hallucinations.

TECHNICAL SKILLS

- **Programming Languages & Web:** Python, SQL, HTML, CSS.
- **Core AI & Vision:** Machine Learning, Deep Learning, Computer Vision, CNN, Explainable AI (SHAP), YOLO, OpenCV.
- **Generative AI & LLMs:** Generative AI, Prompt Engineering, LLM (Applications, Integration, Fine-Tuning), Retrieval-Augmented Generation (RAG).
- **Agents & Frameworks:** Multi-Agent Systems (CrewAI), Agentic AI, LangChain, LangGraph, Vector Databases.
- **Databases & Cloud:** MySQL, SQLite, PostgreSQL, ChromaDB, Supabase, Firebase.
- **DevOps & Tools:** Vercel, Docker, Git, GitHub, CI/CD, GitHub Actions, Linux, Jira, Postman, NGINX.
- **AI Tools & APIs:** Hugging Face Transformers, AI Workflow Automation , OpenAI API , Claude API, Gemini API, n8n.
- **Core CS:** Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms, System Architecture, OOP (Object-Oriented Programming).
- **Productivity Tools:** Word, Excel, PowerPoint, OneNote, Access, Power BI, Power Query, Power Pivot.

CERTIFICATIONS

- Python Certification | FreeCodeCamp.
- Android Development | Great Learning.
- Career Essentials in Project Management | Microsoft.
- Project Management Foundations (PMI) | LinkedIn.
- Career Essentials in Generative AI | Microsoft.

CO-CURRICULAR ACTIVITIES

- Tech Lead – Free and Open Source Software Club (FOSS), SREC.
- Public Relations (PR) Lead – Free and Open Source Software Club (FOSS), SREC.
- Active Member – Computer Society of India (CSI).
- Member – Uyir Club, SREC.